Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | | | | | |

Candidate Number | | | | |

Pearson Edexcel Award

Tuesday 25 June 2024

Afternoon (Time: 3 hours)

Paper reference **9811/01**

Advanced Extension Award Mathematics

Marked Solutions

You must have:
Mathematical Formulae and Statistical Tables
An insert for questions 5, 6 and 7 (enclosed)

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for **each** question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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Pearson

1. In the binomial expansion of

$$(1-8x)^p \qquad |x| < \frac{1}{8}$$

where p is a positive constant,

- the sum of the coefficient of x and the coefficient of x^2 is equal to the coefficient of x^3
- the coefficient of x^2 is positive

Determine the value of p.

(7)

Terms of expansion | Coefficients required

1

$p(-8x)$ $\qquad -8p$

$\frac{p(p-1)}{2}(-8x)^2$ $\qquad \frac{p(p-1)}{2} \times 64$

$\frac{p(p-1)(p-2)}{6}(-8x)^3$ B1 $\qquad \frac{p(p-1)(p-2)}{6} \times -512$

$-8p + 32p(p-1) = \frac{-512p(p-1)(p-2)}{6}$ $\div 8$ M1

$-p + 4p(p-1) = \frac{-64p(p-1)(p-2)}{6}$ $\times 6$

$-6p + 24p(p-1) = -64p(p-1)(p-2)$ $\div 2p$

$-3 + 12(p-1) = -32(p-1)(p-2)$ M1

$12p - 15 = -32(p^2 - 3p + 2)$

$12p - 15 = -32p^2 + 96p - 64$

$32p^2 - 84p + 49 = 0$ A1

$(ax - 7)(bx - 7) = 0$

Since the coefficient of p is negative the factors will have a constant term of -7 since $-7 \times -7 = +49$

Question 1 continued

$ab = 32 \qquad -7a - 7b = -84$

Try 1 and 32 $\quad -7 - 7 \times 32 \neq -84$

2 and 16 $\quad -14 - 7 \times 16 = -14 - 112 \neq -84$

4 and 8 $\quad -28 - 56 = -84$ ✓

$32p^2 - 84p + 49 = 0$

$(4x - 7)(8x - 7) = 0$

$x = \frac{7}{4} \qquad x = \frac{7}{8}$ M1

Since coefficient of $x^2 > 0$ $\qquad 32p(p-1) > 0$

$p(p-1) > 0$

If $x = \frac{7}{4} \qquad \frac{7}{4}\left(\frac{3}{4}\right) = \frac{21}{16} > 0$

If $x = \frac{7}{8} \qquad \frac{7}{8}\left(\frac{-1}{8}\right) = \frac{-7}{64} < 0$

So $x = \frac{7}{4}$ M1 A1

(Total for Question 1 is 7 marks)

2.

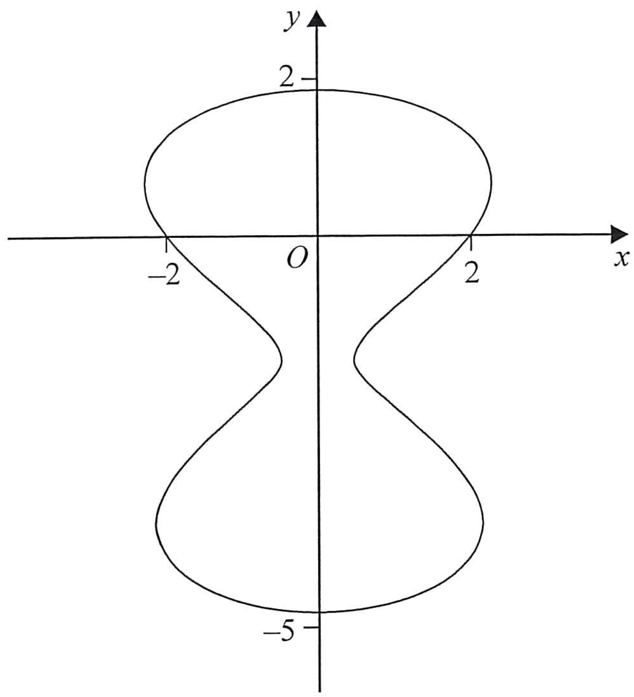

Figure 1

Figure 1 shows the curve defined by the equation

$$y^2 + 3y - 6\sin y = 4 - x^2$$

The point $P(x, y)$ lies on the curve.

The distance from the origin, O, to P is D.

(a) Write down an equation for D^2 in terms of y only.

(1)

(b) Hence determine the minimum value of D giving your answer in simplest form.

(5)

a) $D^2 = x^2 + y^2$

$\downarrow$

$y^2 + 3y - 6\sin y = 4 - x^2 \Rightarrow x^2 = 4 - y^2 - 3y + 6\sin y$

$D^2 = 4 - y^2 - 3y + 6\sin y + y^2$

$D^2 = 4 - 3y + 6\sin y$ B1

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Question 2 continued

b) At the minimum $\frac{dD}{dy} = 0$ so $\frac{dD^2}{dy} = 0$

$D^2 = 4 - 3y + 6\sin y$

$\frac{dD^2}{dy} = -3 + 6\cos y = 0$ M1

$-5 < y < 2$

$6\cos y = 3$

$\cos y = 1/2$

$y = \cos^{-1}(1/2) = \frac{\pi}{3}$ or $-\frac{\pi}{3}$ A1

From the graph the minimum occurs when $y < 0$

$\Rightarrow y = -\pi/3$

$D^2 = 4 - 3\left(-\frac{\pi}{3}\right) + 6\sin\left(-\frac{\pi}{3}\right)$ dM1 A1

$D^2 = 4 + \pi + 6 \times -\frac{\sqrt{3}}{2}$

$D^2 = 4 + \pi - 3\sqrt{3}$

$D = \sqrt{4 + \pi - 3\sqrt{3}}$ A1

(Total for Question 2 is 6 marks)

3. (i) Determine the value of k such that

$$\arctan\frac{1}{2} - \arctan\frac{1}{3} = \arctan k$$

(3)

(ii) (a) Prove that

$$\cos 3A \equiv 4\cos^3 A - 3\cos A$$

(3)

Given that $a = \cos 20°$

(b) write down, in terms of a, an expression for $\cos 40°$

(1)

(c) determine, in terms of a, a simplified expression for $\cos 80°$

(2)

(d) Use part (a) to show that

$$4a^3 - 3a = \frac{1}{2}$$

(1)

(e) Hence, or otherwise, show that

$$\cos 20° \cos 40° \cos 80° = \frac{1}{8}$$

(4)

i) $\arctan(k) = \arctan(1/2) - \arctan(1/3)$

$k = \tan(\arctan(1/2) - \arctan(1/3))$ M1

$\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$ $A = \arctan 1/2$, $B = \arctan 1/3$

$k = \dfrac{\tan\arctan(1/2) - \tan\arctan(1/3)}{1 + \tan\arctan(1/2)\tan\arctan(1/3)}$ M1

$k = \dfrac{1/2 - 1/3}{1 + 1/2 \times 1/3} = \dfrac{1/6}{1 + 1/6} = \dfrac{1}{6} \div \dfrac{7}{6} = \dfrac{1}{6} \times \dfrac{6}{7} = \dfrac{1}{7}$

A1

Question 3 continued

ii a) $\cos(3A) = \cos(2A + A) = \cos(2A)\cos(A) - \sin(2A)\sin(A)$ M1

$\cos(2A) = 2\cos^2(A) - 1$

$= (2\cos^2(A) - 1)\cos(A) - 2\sin(A)\cos(A)\sin(A)$ M1

$\sin(2A) = 2\sin(A)\cos(A)$

$= 2\cos^3(A) - \cos(A) - 2\sin^2(A)\cos(A)$

$= 2\cos^3(A) - \cos(A) - 2(1 - \cos^2(A))\cos(A)$

$= 2\cos^3(A) - \cos(A) - 2\cos(A) + 2\cos^3(A)$

$= 4\cos^3(A) - 3\cos(A)$ A1

ii b) $\cos(40) = \cos(2 \times 20) = 2\cos^2(20) - 1$

$= 2a^2 - 1$ B1

ii c) $\cos(80) = \cos(2 \times 40) = 2\cos^2(40) - 1$

$= 2(2a^2 - 1)^2 - 1$ M1

$= 2(4a^4 - 4a^2 + 1) - 1$

$= 8a^4 - 8a^2 + 1$ A1

ii d) $a = \cos(20)$ from (a)

$4a^3 - 3a = 4\cos^3(20) - 3\cos(20) = \cos(3 \times 20) = \cos(60) = 1/2$ B1

ii e) $\cos(20)\cos(40)\cos(80) = a(2a^2 - 1)(8a^4 - 8a^2 + 1)$ M1

$= (2a^3 - a)(8a^4 - 8a^2 + 1)$

Need to use (a) so rewrite in terms of $4a^3 - 3a$

$2a^3 - a \rightarrow \frac{1}{2}(4a^3 - 3a) + 1/2 a = 1/2 \times 1/2 + 1/2 a = \frac{1}{4} + \frac{1}{2}a$ M1

$8a^4 - 8a^2 + 1 \rightarrow 2a(4a^3 - 3a) - 2a^2 + 1 = 2a \times \frac{1}{2} - 2a^2 + 1$

$= a - 2a^2 + 1$

$(2a^3 - a)(8a^4 - 8a^2 + 1) = (\frac{1}{4} + \frac{1}{2}a)(a - 2a^2 + 1)$ M1

$= \frac{1}{4}(1 + 2a)(a - 2a^2 + 1) = \frac{1}{4}(a - 2a^2 + 1 + 2a^2 - 4a^3 + 2a) = \frac{1}{4}(-4a^3 + 3a + 1)$

$= \frac{1}{4}(1 - (4a^3 - 3a)) = \frac{1}{4}(1 - 1/2) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$ A1

Question 3 continued

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Question 3 continued

(Total for Question 3 is 14 marks)

4. (a) Use the substitution $x = \sqrt{3}\tan u$ to show that

$$\int \frac{1}{3+x^2}\,\mathrm{d}x = p\arctan(px) + c$$

where p is a real constant to be determined and c is an arbitrary constant.

(5)

(b) Use the substitution $x = \dfrac{3u+3}{u-3}$ to determine the exact value of I where

$$I = \int_{-3}^{1} \frac{\ln(3-x)}{3+x^2}\,\mathrm{d}x$$

giving your answer in simplest form.

(10)
(+S1)

B1

a) $x = \sqrt{3}\tan(u)$ $\quad \frac{dx}{du} = \sqrt{3}\sec^2(u)$ $\quad x^2 = 3\tan^2(u)$

$\int \frac{1}{3+x^2}\,dx = \int \frac{1}{3+3\tan^2(u)} \sqrt{3}\sec^2(u)\,du$ M1

$1+\tan^2(u) = \sec^2(u)$
$3+3\tan^2(u) = 3\sec^2(u)$

$= \int \frac{\sqrt{3}\sec^2(u)}{3\sec^2(u)}\,du$ M1

$= \int \frac{\sqrt{3}}{3}\,du = \frac{\sqrt{3}}{3}u + c = \frac{\sqrt{3}}{3}\arctan\left(\frac{x}{\sqrt{3}}\right) + c$

$x = \sqrt{3}\tan(u)$ $\quad = \frac{\sqrt{3}}{3}\arctan\left(\frac{\sqrt{3}x}{3}\right) + c$

$\frac{x}{\sqrt{3}} = \tan(u)$

A1

$\arctan\left(\frac{x}{\sqrt{3}}\right) = u$ B1

Question 4 continued

$$\int_a^b f(x) = -\int_b^a f(x)$$

$$\int_1^{-3} \frac{-\cancel{12}\ln\left(\frac{12}{3-u}\right)}{\cancel{12}(u^2+3)}\,du = \int_{-3}^{1} \frac{\ln\left(\frac{12}{3-u}\right)}{u^2+3}\,du$$ 4th M1

$$= \int_{-3}^{1} \frac{\ln(12)}{u^2+3}\,du - \int_{-3}^{1} \frac{\ln(3-u)}{u^2+3}\,du$$ 3rd M1 A1

$$\searrow I$$

Let original integral $= I$

$$I = \int_{-3}^{1} \frac{\ln(12)}{u^2+3}\,du - I$$

$$2I = \left[\frac{\sqrt{3}}{3}\arctan\left(\frac{\sqrt{3}}{3}u\right)\right]_{-3}^{1} \cdot \ln(12)$$ M1

$$2I = \ln 12\left(\frac{\sqrt{3}}{3}\arctan\left(\frac{\sqrt{3}}{3}\right) - \frac{\sqrt{3}}{3}\arctan(-\sqrt{3})\right)$$ dM1

$$2I = \frac{\sqrt{3}}{3}\ln(12)\left(\frac{\pi}{6}+\frac{\pi}{3}\right) = \frac{\pi\sqrt{3}\ln(12)}{6}$$

$\div 2$

$$I = \frac{\pi\sqrt{3}\ln(12)}{12}$$ A1

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Question 4 continued

b) $$x = \frac{3u+3}{u-3} \qquad \frac{dx}{du} = \frac{(u-3)(3u+3)' - (3u+3)(u-3)'}{(u-3)^2}$$

$$= \frac{3(u-3) - 1(3u+3)}{(u-3)^2} \quad \text{M1}$$

$$= \frac{3u-9-3u-3}{(u-3)^2} = \frac{-12}{(u-3)^2} \quad \text{A1}$$

$$3 - x = 3 - \frac{3u+3}{u-3} = \frac{3(u-3)-(3u+3)}{u-3} = \frac{3u-9-3u-3}{u-3}$$

$$= \frac{-12}{u-3} = \frac{12}{3-u}$$

$$x^2 = \left(\frac{3u+3}{u-3}\right)^2 = \frac{9u^2+18u+9}{(u-3)^2}$$

$$3 + x^2 = 3 + \frac{9u^2+18u+9}{(u-3)^2} = \frac{3(u-3)^2 + 9u^2+18u+9}{(u-3)^2}$$

$$= \frac{3(u^2-6u+9)+9u^2+18u+9}{(u-3)^2} = \frac{12u^2+36}{(u-3)^2}$$

When $x = 1$ $\quad 1 = \frac{3u+3}{u-3} \Rightarrow u-3 = 3u+3$

$-6 = 2u \Rightarrow u = -3$

When $x = -3$ $\quad -3 = \frac{3u+3}{u-3} \Rightarrow -3(u-3) = 3u+3$

$-3u+9 = 3u+3$

$6 = 6u \Rightarrow u = 1$ B1

$$\int_{-3}^{1} \frac{\ln(3-x)}{3+x^2}\,dx = \int_{1}^{-3} \frac{\ln\left(\frac{12}{3-u}\right)}{\frac{12u^2+36}{(u-3)^2}} \cdot \frac{-12}{(u-3)^2}\,du \quad \text{M1}$$

$$= \int_{1}^{-3} \frac{\ln\left(\frac{12}{3-u}\right)}{12u^2+36} \cancel{(u-3)^2} \cdot \frac{-12}{\cancel{(u-3)^2}}\,du$$

Question 4 continued

(Total for Question 4 is 16 marks)

5.

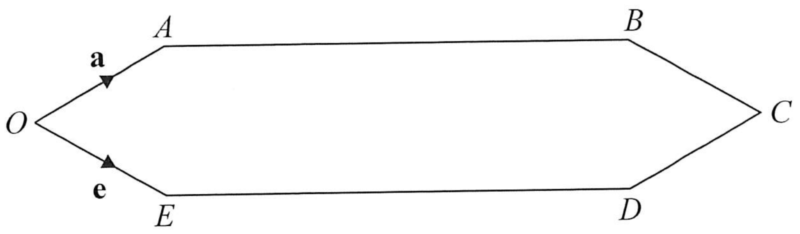


Figure 2

Figure 2 shows a sketch of a hexagon $OABCDE$ where

- the interior angle at O and at C are each 60°
- the interior angle at each of the other vertices is 150°
- $OA = OE = BC = CD$
- $AB = ED = 3 \times OA$

Given that $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OE} = \mathbf{e}$

determine as simplified expressions in terms of $\mathbf{a}$ and $\mathbf{e}$

(a) $\overrightarrow{AB}$ **(3)**

(b) $\overrightarrow{OD}$ **(2)**

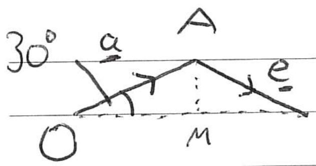
The point R divides AB internally in the ratio $1:2$

(c) Determine $\overrightarrow{RC}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{e}$ **(2)**

The line through the points R and C meets the line through the points O and D at the point X.

(d) Determine $\overrightarrow{OX}$ in the form $\lambda\mathbf{a} + \mu\mathbf{e}$, where λ and μ are real values in simplest form.

(6)
(+S2)



$\overrightarrow{AB} = \lambda(\underline{a} + \underline{e})$ B1

$|OM| = |\underline{a}| \cos(30) = \frac{\sqrt{3}}{2}|\underline{a}|$

$|\underline{a} + \underline{e}| = \frac{\sqrt{3}}{2}|\underline{a}| \times 2 = \sqrt{3}|\underline{a}|$

M1

Since $|AB| = 3|\underline{a}|$ $\quad |AB| = \sqrt{3}(\sqrt{3}|\underline{a}|) = \sqrt{3}|\underline{a} + \underline{e}|$

$\Rightarrow \overrightarrow{AB} = \sqrt{3}(\underline{a} + \underline{e})$ A1

Question 5 continued

M1

b) $\overrightarrow{OD} = \overrightarrow{OE} + \overrightarrow{ED} = \overrightarrow{OE} + \overrightarrow{AB} = \underline{e} + \sqrt{3}(\underline{a} + \underline{e})$ A1

$= \sqrt{3}\,\underline{a} + (1+\sqrt{3})\underline{e}$

c) $AR : RB \Rightarrow \overrightarrow{RB} = \frac{2}{3}\overrightarrow{AB}$

$1 : 2$

$\overrightarrow{RC} = \overrightarrow{RB} + \overrightarrow{BC} = \frac{2}{3}\overrightarrow{AB} + \overrightarrow{BC} = \frac{2}{3}\overrightarrow{AB} + \overrightarrow{OE}$ M1

$= \frac{2}{3}\sqrt{3}(\underline{a}+\underline{e}) + \underline{e} = \frac{2\sqrt{3}}{3}\underline{a} + \left(\frac{2\sqrt{3}}{3}+1\right)\underline{e}$

A1

d) $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AR} + \mu\overrightarrow{RC}$

$= \underline{a} + \frac{\sqrt{3}}{3}(\underline{a}+\underline{e}) + \mu\frac{2\sqrt{3}}{3}\underline{a} + \mu\left(\frac{2\sqrt{3}}{3}+1\right)\underline{e}$ M1 A1

$\overrightarrow{OX} = \gamma\overrightarrow{OD} = \gamma\sqrt{3}\,\underline{a} + \gamma(1+\sqrt{3})\underline{e}$ B1

Equate $\underline{a}$ $\quad 1 + \frac{\sqrt{3}}{3} + \mu\frac{2\sqrt{3}}{3} = \gamma\sqrt{3}$

$\div\sqrt{3}$

$\frac{\sqrt{3}}{3} + \frac{1}{3} + \frac{2}{3}\mu = \gamma$

Equate $\underline{e}$ $\quad \frac{\sqrt{3}}{3} + \mu\left(\frac{2\sqrt{3}}{3}+1\right) = \gamma(1+\sqrt{3})$

$\frac{\sqrt{3}}{3} + \mu\left(\frac{2\sqrt{3}}{3}+1\right) = \left(\frac{\sqrt{3}}{3}+\frac{1}{3}+\frac{2}{3}\mu\right)(1+\sqrt{3})$

$\times 3$

$\sqrt{3} + \mu(2\sqrt{3}+3) = (\sqrt{3}+1+2\mu)(1+\sqrt{3})$

$\sqrt{3} + \mu(2\sqrt{3}+3) = \sqrt{3}+1+2\mu+3+\sqrt{3}+2\sqrt{3}\mu$

$\mu(2\sqrt{3}+3-2-2\sqrt{3}) = \sqrt{3}+1+3+\sqrt{3}-\sqrt{3}$

$\mu = 4+\sqrt{3}$ M1 A1

$\overrightarrow{OX} = \underline{a} + \frac{\sqrt{3}}{3}(\underline{a}+\underline{e}) + (4+\sqrt{3})\frac{2\sqrt{3}}{3}\underline{a} + (4+\sqrt{3})\left(\frac{2\sqrt{3}}{3}+1\right)\underline{e}$

$= \underline{a} + \frac{\sqrt{3}}{3}\underline{a} + \frac{\sqrt{3}}{3}\underline{e} + \frac{8\sqrt{3}}{3}\underline{a} + 2\underline{a} + \frac{8\sqrt{3}}{3}\underline{e} + 4\underline{e} + 2\underline{e} + \sqrt{3}\underline{e}$

$= (3+3\sqrt{3})\underline{a} + (6+4\sqrt{3})\underline{e}$ A1

Question 5 continued

Question 5 continued

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 15 marks)

6.

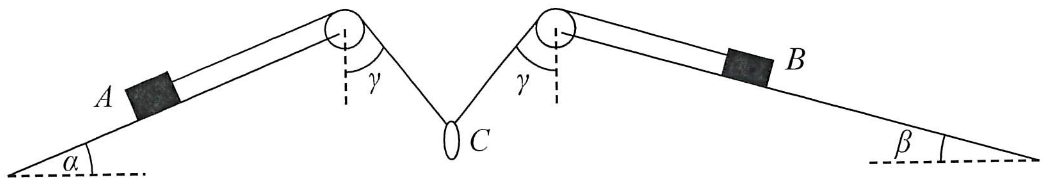


Figure 3

Figure 3 shows a block A with mass $4m$ and a block B with mass $5m$.
Block A is at rest on a rough plane inclined at an angle α to the horizontal.
Block B is at rest on a rough plane inclined at an angle β to the horizontal.

The blocks are connected by a light inextensible string which passes over a small smooth pulley at the top of each plane.

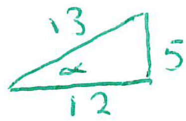
A small smooth ring C, of mass $8m$, is threaded on the string between the pulleys so that A, B and C all lie in the same vertical plane.

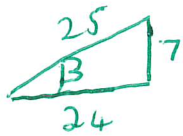
The part of the string between A and its pulley lies along a line of greatest slope of the plane of angle α.

The part of the string between B and its pulley lies along a line of greatest slope of the plane of angle β.

The angle between the vertical and the string between each pulley and the ring C is γ.

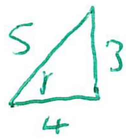
The two blocks, A and B, are modelled as particles.





Given that

- $\tan\alpha = \dfrac{5}{12}$ and $\tan\beta = \dfrac{7}{24}$ and $\tan\gamma = \dfrac{3}{4}$

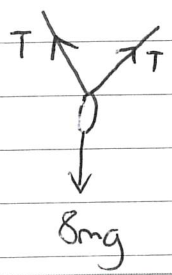


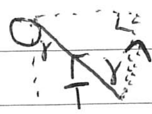
- the coefficient of friction, μ, is the same between each block and its plane
- one of the blocks is on the point of sliding up its plane
- the tension in the string is T

(a) determine, in terms of m and g, an expression for T, **(3)**

(b) draw a diagram showing the forces on block A, clearly labelling each of the forces acting on the block, **(3)**

(c) determine the value of μ, giving a justification for your answer. **(10)**
(+S2)





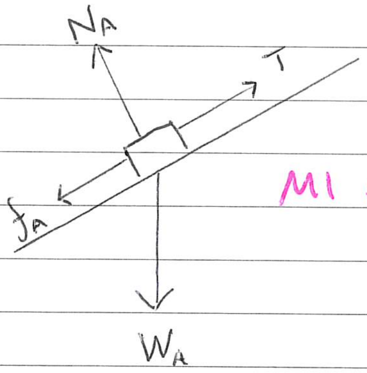
$T\cos\gamma = T \times \frac{4}{5}$

$\frac{4}{5}T + \frac{4}{5}T = 8mg \Rightarrow \frac{8T}{5} = 8mg \Rightarrow T = 5mg$

M1 M1 A1

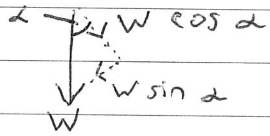
Question 6 continued

b)



M1 A1 A1

N_A is the normal reaction
T is the tension in the string
W_A is the weight of block A
f_A is the friction between the slope and block A.



c) Block A: parallel to plane $T - f_A - 4mg\sin\alpha = m \times 0$

$5mg - 4mg(5/13) = f_A$ 2nd M1

$\frac{45}{13}mg = f_A$

perpendicular to plane $N_A = 4mg\cos\alpha = 4mg(12/13)$ M1 A1

$N_A = \frac{48}{13}mg$

One block is on the point of sliding so for one block $F = \mu R$

$\mu_A = \frac{f}{R} = \frac{45mg}{13} \div \frac{48}{13}mg = \frac{45}{48} = \frac{15}{16}$ M1 A1

for block B the equations would be of the same form but with $W = 5mg$ and angle β $\therefore$ $f_B = 5mg - 5mg(7/25) = \frac{18}{5}mg$

$N_B = 5mg(24/25) = \frac{24}{5}mg$ M1
B1

$\mu_B = \frac{18}{5}mg \div \frac{24}{5}mg = \frac{18}{24} = \frac{3}{4}$ A1

Since $\frac{3}{4} < \frac{15}{16}$ $\mu = \frac{15}{16}$ M1 A1

If $\mu = \frac{3}{4}$ $\mu N_A = \frac{36}{13}mg$ so $f_A < \mu N_A$ hence not at maximum and thus not on the point of sliding, the system would not move.

Question 6 continued

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Question 6 continued

Question 6 continued

Question 6 continued

(Total for Question 6 is 18 marks)

7.

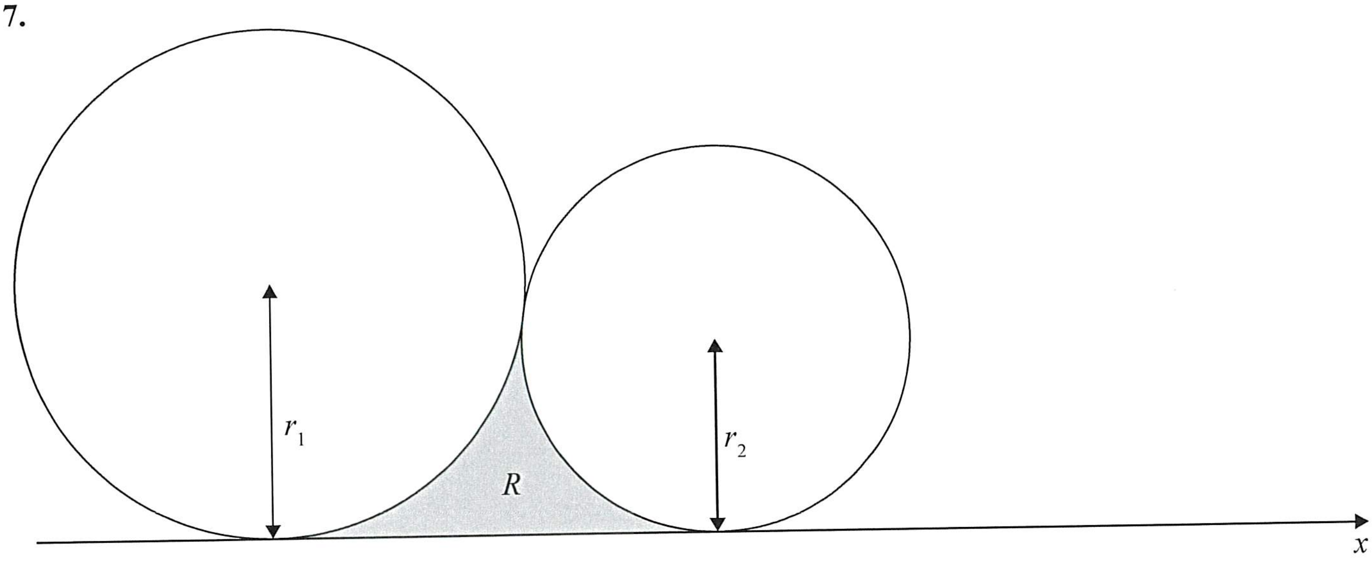


Figure 4

Figure 4 shows a circle with radius r_1 and a circle with radius r_2
The circles touch externally at a single point above the x-axis.
Both circles also have the x-axis as a tangent.

(a) Show that the horizontal distance between the centres of the circles, d, is given by

$$d^2 = 4r_1r_2$$

(2)

The finite region R, shown shaded in Figure 4, is bounded by the x-axis and minor arcs of the two circles.

Given that $r_1 \geqslant r_2$

(b) show that the area of R is given by

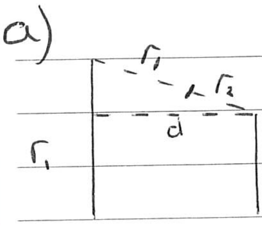
$$(r_1 + r_2)\sqrt{r_1 r_2} - \frac{1}{2}(r_1^2 - r_2^2)\theta - \frac{1}{2}\pi r_2^2$$

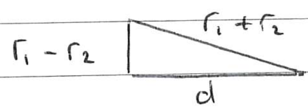
where $\cos\theta = \dfrac{r_1 - r_2}{r_1 + r_2}$

(5)

Question 7 continues on the next page.

a)





$d^2 + (r_1 - r_2)^2 = (r_1 + r_2)^2$ M1

$d^2 + r_1^2 - 2r_1r_2 + r_2^2 = r_1^2 + 2r_1r_2 + r_2^2$

$d^2 = 4r_1r_2$ A1*

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Question 7 continued

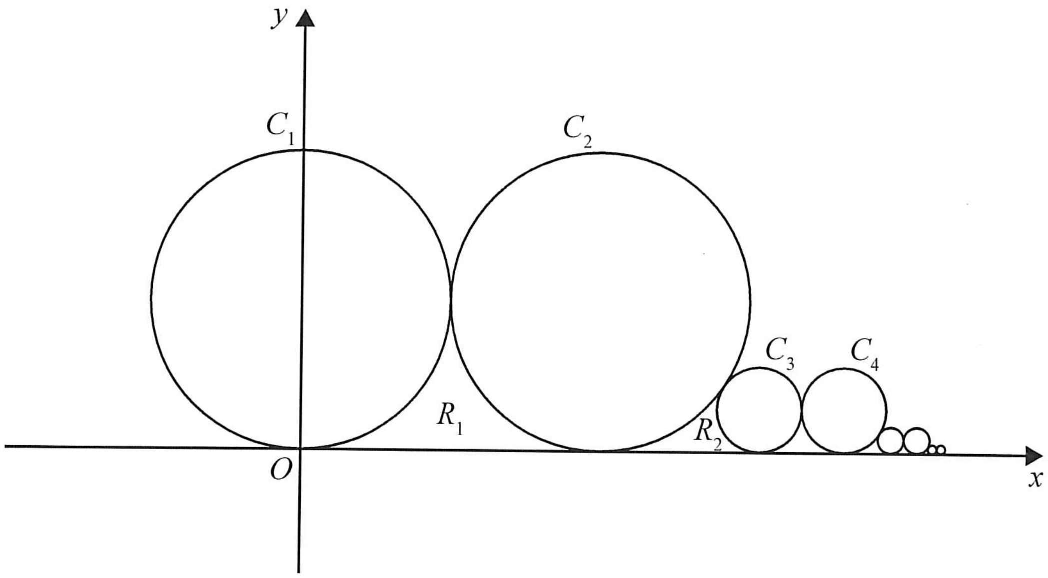


Figure 5

A sequence of circles, C_1, C_2, C_3, ... with radii r_1, r_2, r_3, ... respectively, is constructed such that

- each circle is tangential to and above the x-axis
- the first circle, C_1, has centre (0, 1)
- each successive circle touches the preceding one externally at a single point
- the horizontal distances between the centres of successive circles form a geometric sequence with first term 2 and common ratio $\frac{1}{\sqrt{3}}$

The first few circles in the sequence are shown in Figure 5.

(c) (i) Determine the value of r_3

(ii) Show that, for $n \geqslant 1$, $r_{n+2} = kr_n$ where k is a constant to be determined.

(iii) Hence show that, for $n \geqslant 1$, $r_{2n} = r_{2n-1}$

(9)

The region bounded between C_n, C_{n+1} and the x-axis is R_n

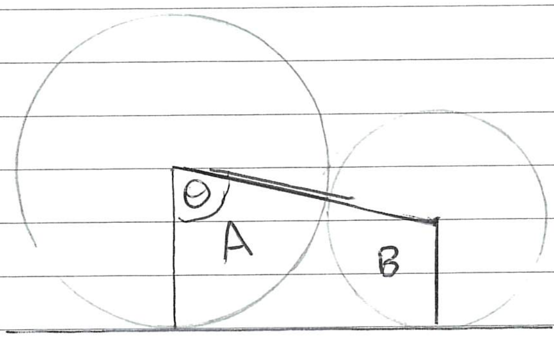
The total area, A, bounded above the x-axis and under all the circles is the sum of the areas of all these regions.

(d) Determine the value of A, giving the answer in simplest form.

(6)
(+S2)

Question 7 continued

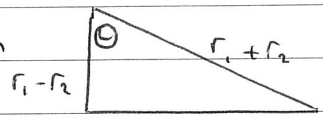
b)



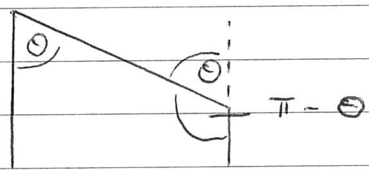
$R = \text{trapezium} - \text{sector } A - \text{Sector } B$

If $\cos\theta = \dfrac{r_1 - r_2}{r_1 + r_2} = \dfrac{adj}{hyp}$

then



B1



$\text{trapezium} = \dfrac{r_1 + r_2}{2} \times d = \dfrac{r_1 + r_2}{2} \times \sqrt{4r_1r_2} = (r_1 + r_2)\sqrt{r_1r_2}$ 2nd M1

$\text{Sector } A = \dfrac{1}{2} r_1^2 \theta$ $\qquad$ $\text{Sector } B = \dfrac{1}{2} r_2^2 (\pi - \theta)$ M1

$R = (r_1 + r_2)\sqrt{r_1r_2} - \dfrac{1}{2}r_1^2\theta - \left(\dfrac{1}{2}r_2^2\pi - \dfrac{1}{2}r_2^2\theta\right)$ M1

$= (r_1 + r_2)\sqrt{r_1r_2} - \dfrac{1}{2}r_1^2\theta + \dfrac{1}{2}r_2^2\theta - \dfrac{1}{2}r_2^2\pi$

$= (r_1 + r_2)\sqrt{r_1r_2} - \dfrac{1}{2}(r_1^2 - r_2^2)\theta - \dfrac{1}{2}\pi r_2^2$ A1*

c i) distance between C_2 and $C_1 = 2$ $\quad$ Since $r_1 = 1$, $r_2 = 1$

distance between C_3 and $C_2 = \dfrac{2}{\sqrt{3}} = \dfrac{2\sqrt{3}}{3}$

from (a) $d^2 = 4r_n r_{n+1}$

$\left(\dfrac{2\sqrt{3}}{3}\right)^2 = 4r_2r_3 \Rightarrow \dfrac{4}{3} = 4 \times 1 \times r_3 \Rightarrow \dfrac{1}{3} = r_3$

M1 A1

Question 7 continued

c ii) Let distance between C_n and C_{n+1} be d_n
then distance between C_{n+1} and C_{n+2} is d_{n+1}

$$d_{n+1} = \frac{1}{\sqrt{3}} d_n$$

$$(d_{n+1})^2 = \left(\frac{1}{\sqrt{3}} d_n\right)^2 = \frac{1}{3} d_n^2 \quad M1$$

$$4 r_{n+1} r_{n+2} = \frac{1}{3} \cdot 4 r_n r_{n+1} \quad M1$$

$\div 4 r_{n+1}$

$$r_{n+2} = \frac{1}{3} r_n \quad M1 \; A1$$

c iii) $r_1 = 1 \quad r_3 = \frac{1}{3} \quad r_5 = \frac{1}{9} \quad r_7 = \frac{1}{27} \; \ldots$ from (c ii)

Likewise $r_2 = 1 \quad r_4 = \frac{1}{3} \quad r_6 = \frac{1}{9} \; \ldots$

Each form a geometric sequence with $a = 1 \quad r = 1/3$

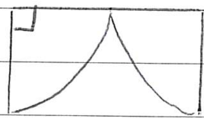
$$r_{odd} = r_{2n-1} = 1 \times \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^{n-1} \quad M1$$

$$r_{even} = r_{2n} = 1 \times \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^{n-1} \quad M1 \qquad \therefore r_{2n} = r_{2n-1} \quad A1$$

$n \geq 1$

d) For areas R_1 R_3 R_5 etc the adjacent circles have equal radii

$$R_{2n-1} \quad r = \left(\frac{1}{3}\right)^{n-1} \quad M1$$



$\theta = \frac{\pi}{2}$

from (b)

$$R_{2n-1} = \left(\left(\frac{1}{3}\right)^{n-1} \times 2\right) \sqrt{\left(\left(\frac{1}{3}\right)^{n-1}\right)^2} - 0 - \frac{1}{2} \pi \left(\left(\frac{1}{3}\right)^{n-1}\right)^2$$

$$= \left(\frac{1}{3}\right)^{n-1} \left(2 \left(\frac{1}{3}\right)^{n-1} - \frac{1}{2} \pi \left(\frac{1}{3}\right)^{n-1}\right)$$

$$= \left(\left(\frac{1}{3}\right)^{n-1}\right)^2 \left(2 - \frac{\pi}{2}\right)$$

$$= \left(\frac{1}{9}\right)^{n-1} \left(2 - \frac{\pi}{2}\right)$$

Question 7 continued

For areas R_2 R_4 R_6 etc. the adjacent circles have different radii.

For R_{2n} radius of larger circle $= \left(\frac{1}{3}\right)^{n-1}$

radius of smaller circle $= \frac{1}{3} \times \left(\frac{1}{3}\right)^{n-1}$

from (b) $\cos\theta = \frac{r_1 - r_2}{r_1 + r_2}$

$$\cos\theta = \frac{\left(\frac{1}{3}\right)^{n-1} - \frac{1}{3}\times\left(\frac{1}{3}\right)^{n-1}}{\left(\frac{1}{3}\right)^{n-1} + \frac{1}{3}\times\left(\frac{1}{3}\right)^{n-1}} = \frac{\frac{2}{3}\left(\frac{1}{3}\right)^{n-1}}{\frac{4}{3}\left(\frac{1}{3}\right)^{n-1}} = \frac{2}{3} \div \frac{4}{3} = \frac{2}{4} = \frac{1}{2}$$

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) \quad \theta = \frac{\pi}{3}$$ B1

from (b)

$$\text{Area } R_{2n} = \left(\left(\frac{1}{3}\right)^{n-1} + \frac{1}{3}\times\left(\frac{1}{3}\right)^{n-1}\right)\sqrt{\frac{1}{3}\times\left(\left(\frac{1}{3}\right)^{n-1}\right)^2} - \frac{1}{2}\left(\left(\left(\frac{1}{3}\right)^{n-1}\right)^2 - \left(\frac{1}{3}\times\left(\frac{1}{3}\right)^{n-1}\right)^2\right)\frac{\pi}{3}$$
$$- \frac{\pi}{2}\left(\frac{1}{3}\times\left(\frac{1}{3}\right)^{n-1}\right)^2$$ M1

$$= \frac{4}{3}\times\left(\frac{1}{3}\right)^{n-1}\times\sqrt{\frac{1}{3}}\times\left(\frac{1}{3}\right)^{n-1} - \frac{\pi}{6}\left(\left(\frac{1}{9}\right)^{n-1} - \frac{1}{9}\times\left(\frac{1}{9}\right)^{n-1}\right) - \frac{\pi}{2}\times\frac{1}{9}\times\left(\frac{1}{9}\right)^{n-1}$$

$$= \frac{4\sqrt{3}}{9}\times\left(\frac{1}{9}\right)^{n-1} - \frac{\pi}{6}\times\frac{8}{9}\times\left(\frac{1}{9}\right)^{n-1} - \frac{\pi}{18}\times\left(\frac{1}{9}\right)^{n-1}$$

$$= \left(\frac{1}{9}\right)^{n-1}\left(\frac{4\sqrt{3}}{9} - \frac{8\pi}{54} - \frac{\pi}{18}\right) = \left(\frac{1}{9}\right)^{n-1}\left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}\right)$$

Question 7 continued

For R_1 R_{2n-1} $n=1$ Area $= 2 - \pi/2$

R_3 R_{2n-1} $n=2$ Area $= \frac{1}{9}(2 - \pi/2)$

R_5 R_{2n-1} $n=3$ Area $= \frac{1}{9^2}(2 - \pi/2)$

Geometric series $a = 2 - \pi/2$ $r = 1/9$

M1

$$S_\infty = \frac{2 - \pi/2}{1 - 1/9} = \left(2 - \frac{\pi}{2}\right) \div \frac{8}{9} = \left(2 - \frac{\pi}{2}\right) \times \frac{9}{8}$$

For R_2 R_{2n} $n=1$ Area $= \frac{4\sqrt{3}}{9} - \frac{11\pi}{54}$

R_4 R_{2n} $n=2$ Area $= \frac{1}{9}\left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}\right)$

Again geometric series $a = \frac{4\sqrt{3}}{9} - \frac{11\pi}{54}$ $r = \frac{1}{9}$

$$S_\infty = \frac{\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}}{1 - 1/9} = \left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}\right) \div \frac{8}{9} = \left(\frac{4\sqrt{3}}{9} - \frac{11\pi}{54}\right) \times \frac{9}{8}$$

dM1

Total area $= \frac{9}{8}\left(2 - \frac{\pi}{2} + \frac{4\sqrt{3}}{9} - \frac{11\pi}{54}\right) = \frac{9}{8}\left(2 + \frac{4\sqrt{3}}{9} - \frac{38\pi}{54}\right)$

$= \frac{9}{4} + \frac{\sqrt{3}}{2} - \frac{19\pi}{24}$ A1

Question 7 continued

(Total for Question 7 is 24 marks)

TOTAL FOR PAPER IS 100 MARKS